SMALL-SCALE PIPING EXPERIMENTS IN HETEROGENEOUS SANDS

Giacomo Negrinelli [1], Vera M. Van Beek [2] & Roberto Ranzi [3]

(1) Studio Frosio S.r.l. (previously Università degli Studi di Brescia); (2) Department of Geo-engineering, Deltares; (3) DICATAM, Università degli Studi di Brescia

KEY POINTS

- *Backward erosion piping is a dominant failure mechanism for levees that determines uncertainty in residual flooding hazard and risk mapping*
- *The occurrence of piping through the levee foundation can be predicted using the Sellmeijer model, which is developed and validated only for homogeneous granular layers*
- *Small-scale laboratory experiments indicated a substantial increase of piping resistance in heterogeneous sand samples with micro- and macro-scale variation of properties in the pipe path*

1 INTRODUCTION

Backward erosion piping is an internal erosion mechanism whereby shallow pipes are formed in the granular foundation of a water-retaining structure due to the removal of sand particles under the action of the seepage flow. Backward erosion piping is one of the most dominant failure mechanisms for river levees (*Colleselli,* 1994). Several levee failure events have been attributed to piping (*Yao et al.,* 2009; *Vrijling,* 2010; *Calabresi et al.,* 2013), determining uncertainty in residual flooding hazard and risk mapping (*Mazzoleni et al.,* 2014).

The occurrence of piping through the levee foundation can be predicted, for instance, using the Sellmeijer model (*Sellmeijer et al.,* 2011). The Sellmeijer model can estimate the critical hydraulic head across the levee that causes the complete development of the pipe leading to failure. The Sellmeijer model is developed and validated only for homogeneous granular layers. However, the subsurface encountered below levees can be far from homogeneous because variation in soil characteristics from micro-scale to macro-scale is common in the field due to the complexity of the geology of the shallow subsurface of river systems.

Previous laboratory tests (*Van Beek et al.,* 2008; *Van Beek et al.,* 2015a) have been performed using sand samples characterized by variation of properties in the path of the pipe like grain size and relative density. They have indicated a substantial increase of piping resistance of the sandy aquifer, which has not been predicted by the Sellmeijer model. In this paper small-scale experiments carried out at the Deltares Geo Model Laboratory are described in order to have insight in the effect of micro-scale and macro-scale heterogeneity on progression of pipe development.

2 METHOD

The process of backward erosion piping in heterogeneous sand samples was investigated by means of small-scale experiments with circular outlet configuration, which have been described in detail in *Van Beek*

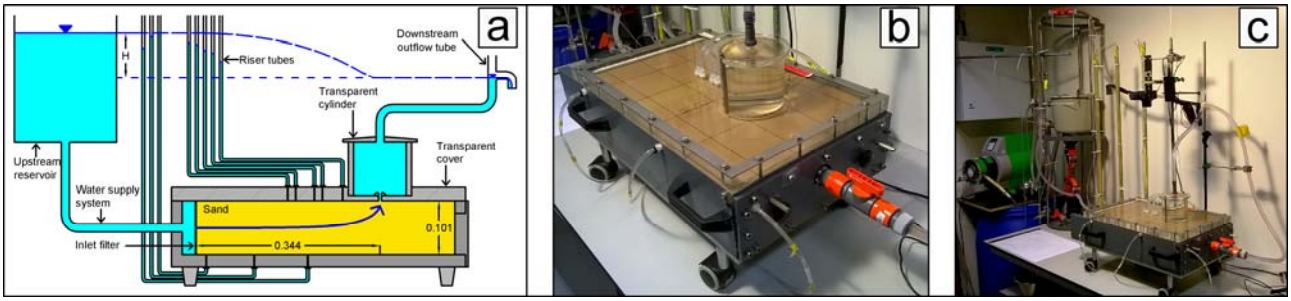


Figure 1. Small-scale experimental setup.

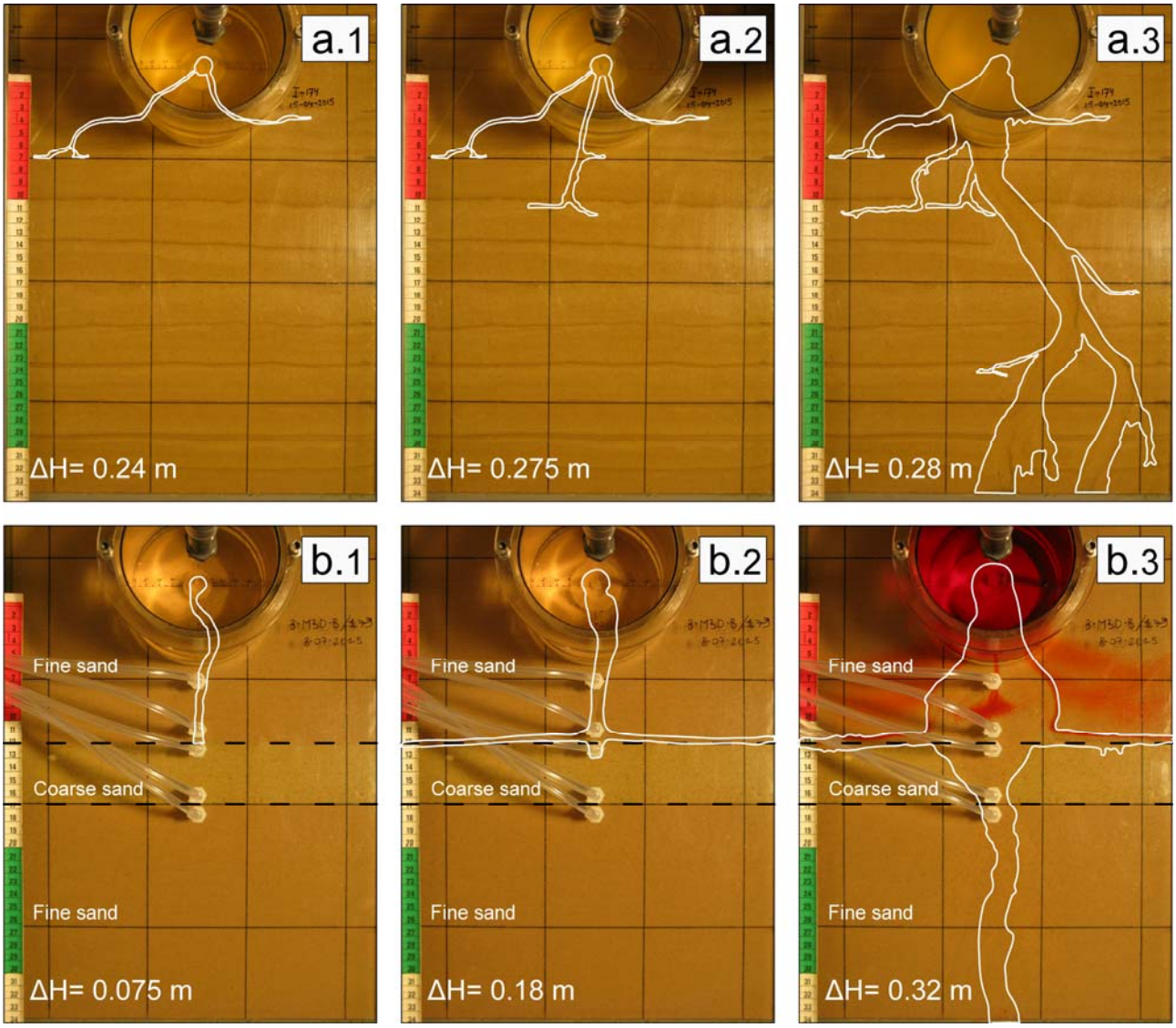


Figure 2. Area of the sample affected by piping at different stages of pipe development for a micro-scale (a) and a macro-scale (b) heterogeneity experiment (respectively, test I-174 and B-M3D-B-179).

et al. (2015a). The experiments were performed using a setup consisting of a rigid box filled with sand, a water supply system and several riser tubes to measure pore pressure at various locations in the sand sample (Fig. 1). The sand box was subjected to a horizontal hydraulic gradient and during the test the hydraulic head over the sand sample was raised stepwise until the process of piping was observed through a transparent cover. All the experiments were prepared using the wet method described by *Van Beek et al.* (2015a) and the preparation technique described by *De Lange et al.* (2016) in order to obtain a fully saturated and dense (bulk relative density RD > 85%) heterogeneous sand sample.

Two series of experiments were performed. In the first series the effect of micro-scale heterogeneity on piping was investigated. In the field micro-scale variations are naturally present resulting in local differences of sand properties in the path of the pipe. These variations were reproduced in laboratory preparing a sand sample in which a sequence of layers with small variation in sand properties was created perpendicular to the direction of the seepage flow. One test was performed using Enschedé sand (d_{70}= 0.431 mm and d_{60}/d_{10}= 1.6) while two other tests were performed using Itterbeck mixture 2 sand (d_{70}= 0.203 mm and d_{60}/d_{10}= 3.2).

In the second series the objective was to investigate the effect of macro-scale heterogeneity on the progression of pipe development. In the field, macro-scale heterogeneity can be caused by the presence of coarse granular lenses in the levee subsurface or it may also correspond to the situation in which a coarse

Exp. no.	Type of heterogeneity	Sand type		Permeability: m/s		RD		i_{crit}
		Fine sand	Coarse sand	K_{fine}	K_{coarse}	RD_{fine}	RD_{coarse}	
E-173	Micro-scale	Enschedé sand		4.83E-04		0.90		0.33
I-174	Micro-scale	Itterbeck mixture 2		2.49E-05		0.85		0.81
I-175	Micro-scale	Itterbeck mixture 2		2.15E-05		0.87		1.19
B-SZ-B-176	Macro-scale	Baskarp	Itterbeck 431 µm	6.07E-05	1.03E-04	0.92	0.90	0.80
B-SZ-B-177	Macro-scale	Baskarp	Itterbeck 431 µm	5.81E-05	1.00E-04	0.97	0.87	1.08
B-SZ-B-178	Macro-scale	Baskarp	Itterbeck 431 µm	5.85E-05	1.03E-04	0.90	0.95	1.16
B-M3D-B-179	Macro-scale	Baskarp	Itterbeck 333 µm	6.13E-05	1.19E-04	0.90	0.95	0.93
B-M3D-B-180	Macro-scale	Baskarp	Itterbeck 333 µm	5.56E-05	1.19E-04	0.90	0.95	1.42

Table 1. Overview of experimental results.

sand barrier is installed underneath a levee as a piping prevention measures (*Van Beek et al.,* 2015b). Macro-scale heterogeneity was reproduced preparing a sample consisting of homogeneous fine sand with an intermediate zone of homogeneous coarse sand. Fine Baskarp sand (d_{70}= 0.140 mm and d_{60}/d_{10}= 1.5) was used in all the tests. Three tests were performed using an intermediate zone of medium coarse Itterbeck 431 µm sand (d_{70}= 0.505 mm and d_{60}/d_{10}= 2.6) while in two other tests the intermediate zone was made of medium coarse Itterbeck 333 µm sand (d_{70}= 0.349 mm and d_{60}/d_{10}= 2.1). As suggested by *Koelewijn et al.* (2014), filter criteria were adopted in order to verify for each experiment with macro-scale heterogeneity if the intermediate coarse sand layer had sufficient permeability, if it could retain the finer fraction of the sand sample and if it was internally stable. The first two criteria have been described by Terzaghi and given by *CUR* (2010) as $d_{15C} \geq 5d_{15F}$ and $d_{15C} \leq 5d_{85F}$, respectively, where the subscript C stands for the coarse sand of the intermediate layer and the subscript F for fine sand. Internal stability has been investigated by *Burenkova* (1993), whose findings can be formulated as the inequality:

$$0.76\log\left(\frac{d_{90}}{d_{15}}\right)+1<\left(\frac{d_{90}}{d_{60}}\right)<1.86\log\left(\frac{d_{90}}{d_{15}}\right)+1 \quad (1)$$

While for all the tests the sand sample met the retention and internal stability criteria, the permeability criterion was not satisfied and for this reason, according to *CUR* (2010), progressive accumulation of fines was likely to be expected in the pores of the coarse sand layer at the upstream interface.

3 RESULTS

Overview of the experiments is presented in Table 1. Photos of different stages of pipe development during the experiments are shown in Figure 2.

Photos show that progression of pipe development was influenced by micro-scale heterogeneity. It was found that experiments with a layered sand sample resulted in a critical gradient higher than the one of homogeneous equivalent experiments described in *Van Beek et al.* (2015a). It was noted that, in micro-scale heterogeneity experiments on non-uniform sand type, greater layering and low permeability was observed, leading to a larger increase of strength of the sand sample.

Photos show that progression of pipe development was influenced by macro-scale heterogeneity. In each test, pipe progression was interrupted at the downstream interface between fine and coarse sand and, until a significant head increase didn't occur, the pipe developed adjacent to the interface. It was found that in case of heterogeneous experiments the critical head for the complete progression of the pipe was larger compared to the results of homogeneous equivalent tests performed by Van Beek et al. (2015a), although it was noted that at the beginning of the experiments the pipe developed in the downstream fine sand layer at a lower head drop. It was noted that, although a correlation between piping resistance and grain size of the intermediate coarse sand layer was expected, the final critical head was strongly affected by different erosion patterns occurred in experiments with same characteristics.

4 DISCUSSION

The effect of micro-scale heterogeneity on the piping process could be explained considering that, in a layered sand sample, layers of fines with higher relative density led to local decrease of permeability and layers with relatively coarse grains were more difficult to erode, according to *Van Beek et al.* (2015a).

The effect of macro-scale heterogeneity on the piping process could be explained considering that, according to *Van Beek et al.* (2015a), the presence of the intermediate coarse sand layer resulted in a large increase of strength of the sand sample because, when the layer with coarser grains was encountered by the pipe, a larger zone needed to be fluidized upstream the pipe tip. Therefore, lengthening was interrupted and pipe formation occurred in direction perpendicular to the layer interface through fine sand until the overall head was such that the local gradient in the coarse sand upstream the pipe was high enough to cause erosion.

5 CONCLUSION

Small-scale piping experiments on heterogeneous sand samples revealed that variation of sand properties in the path of the pipe caused by micro-scale and macro-scale heterogeneity has a significant influence on the progression of pipe development, increasing the piping resistance of the sandy aquifer.

Further research is required to allow for the modelling of backward erosion piping in heterogeneous sands in order to include the effect of natural variation in soil properties on piping in the safety assessment of water-retaining structures and to develop innovative piping prevention measures.

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